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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING – (24CI3201R)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	**AI-Powered Medical Image Analysis Platform for Early Disease Detection**	
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Abstract

The increasing demand for accurate and timely medical diagnosis has highlighted the significant limitations of traditional manual image analysis methods used in healthcare. Medical professionals are often overwhelmed with large volumes of patient data, leading to delayed diagnoses and increased risk of human error. Misdiagnosis and late detection of diseases such as pneumonia and other chest-related conditions continue to pose serious threats to patient health and safety. This project addresses these challenges by presenting an AI-powered platform specifically designed to assist healthcare professionals in detecting diseases at an early stage through automated medical image analysis, reducing the burden on clinicians while improving the speed and accuracy of diagnosis.

The proposed platform leverages advanced machine learning models trained on chest X-ray datasets to classify and identify potential abnormalities with high accuracy. Built following Agile development principles across multiple structured sprints, the system ensures continuous collaboration, iterative improvements, and timely delivery of features. The platform further incorporates secure development practices at every stage of the software lifecycle, ensuring that patient data is handled responsibly and in compliance with data protection regulations. A robust continuous delivery pipeline automates the process of testing, security scanning, and deploying updates, while cloud infrastructure ensures the platform remains scalable, accessible, and highly available to healthcare institutions of varying sizes.

Beyond development, the platform places strong emphasis on operational excellence and long-term reliability. Real-time monitoring capabilities allow the system to track model performance and detect any degradation in prediction accuracy over time, triggering automated retraining when necessary. Patient data privacy and ethical considerations are embedded throughout the system, with bias auditing mechanisms ensuring the AI model performs fairly across diverse patient demographics. By combining the power of artificial intelligence with modern software engineering and security practices, this platform aims to fundamentally transform the diagnostic process, reduce medical errors, support informed clinical decision-making, and ultimately contribute to better patient health outcomes across healthcare institutions.

Signature of the Guide HOD